

CSE/ISyE 6740
Computational Data Analysis

Density Estimation

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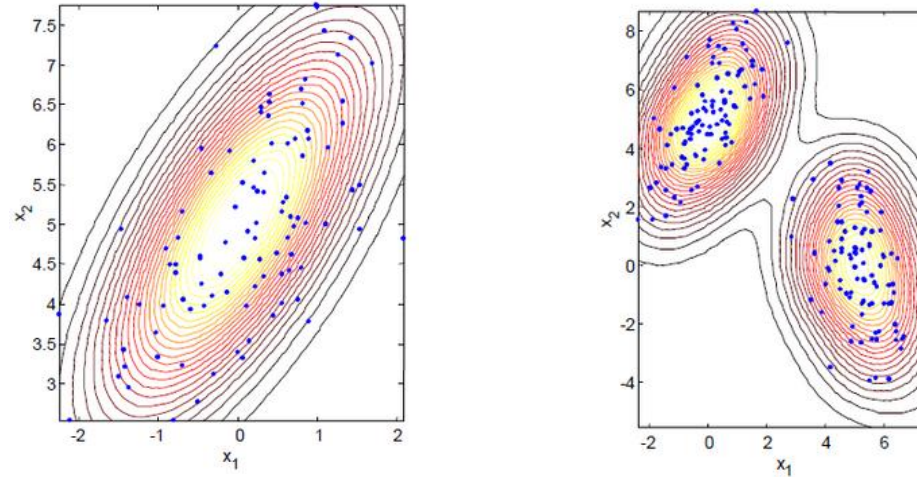
Outline

- **Unsupervised learning**
 - Density estimation
 - Parametric models
 - Non-parametric models
 - Kernel density estimation
 - Gaussian mixture models
 - Expectation-Maximization algorithm

Density Estimation

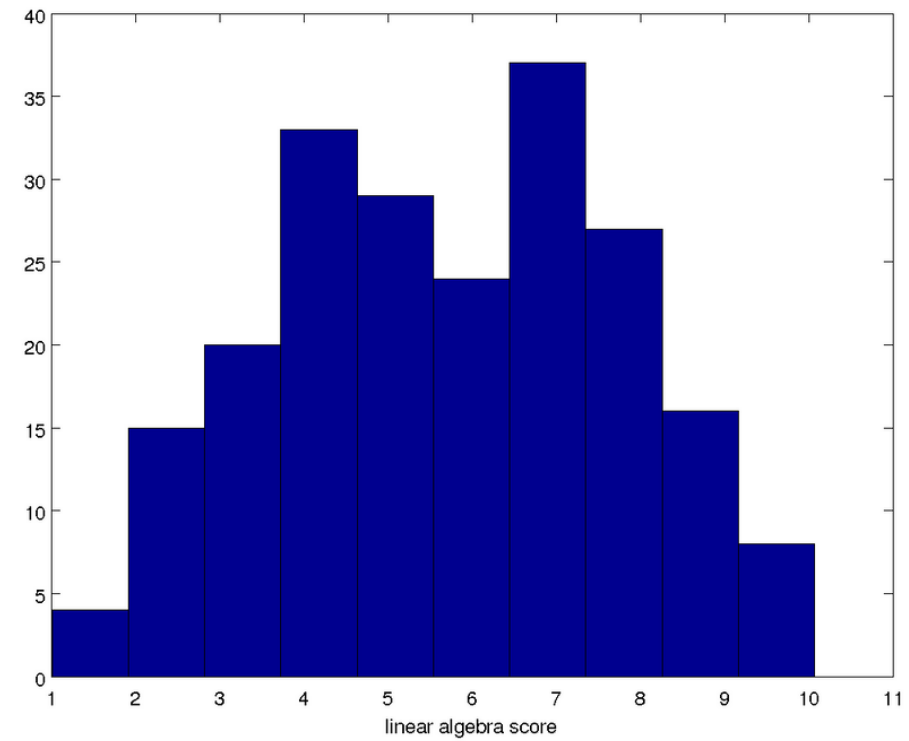
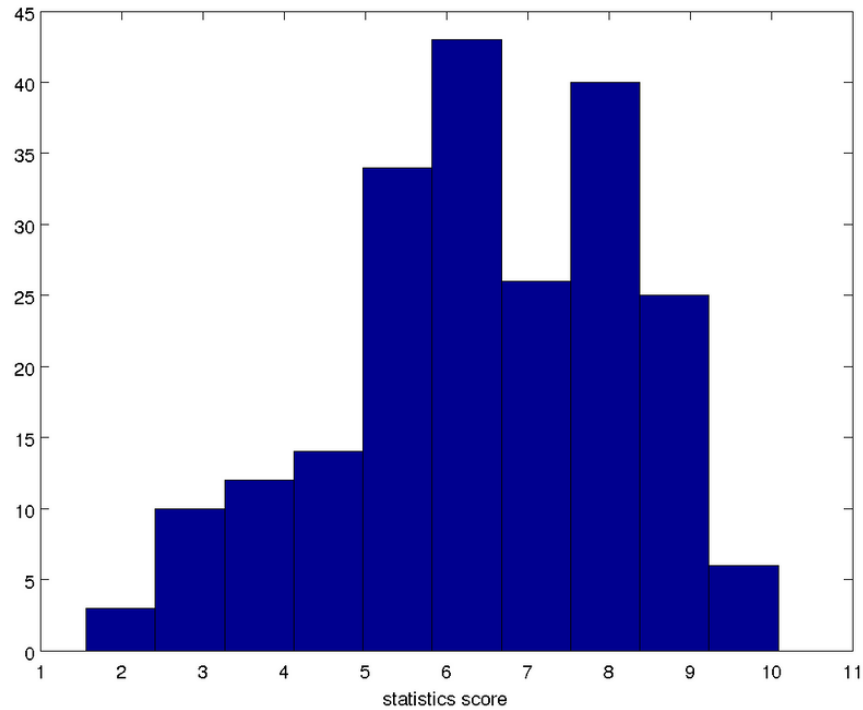
What is Density Estimation?

- Learn more about the “shape” of the data cloud

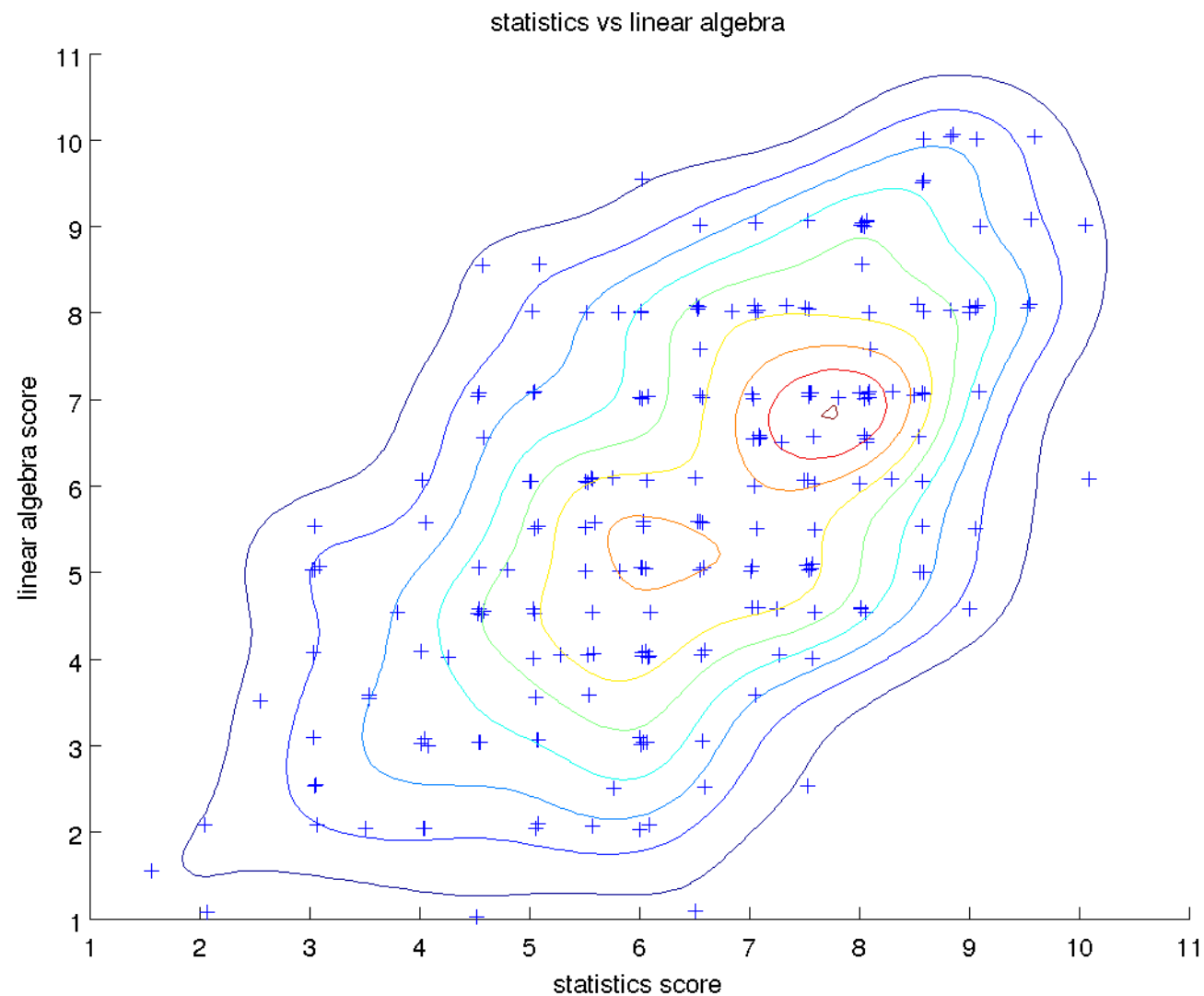


- Assess the likelihood of seeing a particular data point
 - Is this a typical data point? (high density value)
 - Is this an abnormal data point / outlier? (low density value)

Example: Test Scores (conti.)



Example: Test Scores (conti.)



Parametric Models

Parametric Models

- Models which can be described by **a fixed number of parameters**

- Discrete case:** e.g., Bernoulli distribution

$$P(x|\theta) = \theta^x(1 - \theta)^{1-x}$$

- One parameter $\theta \in [0,1]$, which generates a family of models: $\mathcal{F} = \{P(x|\theta) \mid \theta \in [0,1]\}$

- Continuous case:** e.g., multivariate Gaussian distribution

$$P(x|\mu, \Sigma) = \frac{1}{|\Sigma|^{\frac{1}{2}}(2\pi)^{\frac{n}{2}}} \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right)$$

- Two sets of parameters (μ, Σ) , which again generate a family of models: $\mathcal{F} = \{P(x|\mu, \Sigma) \mid \mu \in \mathbb{R}^n, \Sigma \in \mathbb{R}^{n \times n} \text{ and PSD}\}$

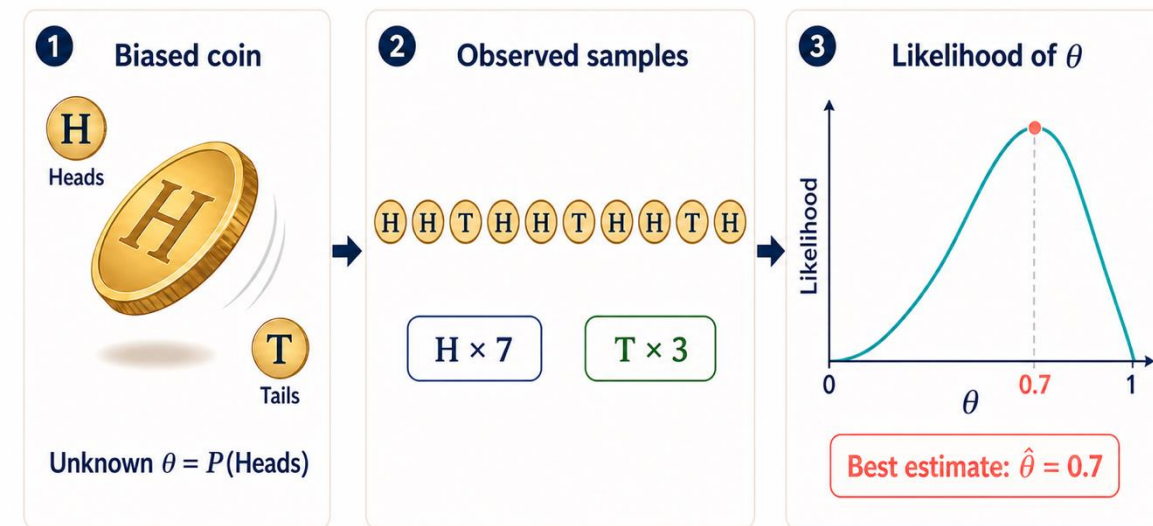
Maximum Likelihood Estimator

- A very popular estimator is the **maximum likelihood estimator (MLE)**, which is simple and has good statistical properties.
- Assume that n data points $D = \{x^1, x^2, \dots, x^n\}$ drawn independently and identically (i.i.d.) from some distribution $P^*(x)$
- Fit the model $P(x|\theta)$ to the dataset D

$$\theta = \operatorname{argmax}_{\theta} \log \prod_{i=1}^n P(x^i|\theta)$$

Example: Biased Coin

- Estimate the probability θ of landing in heads using a biased coin
- Given a sequence of n independently and identically distributed (i.i.d.) flips
 - E.g., $D = \{x^1, x^2, \dots, x^n\}$ (e.g., $\{1, 0, 1, \dots, 0\}$). where $x^i \in \{0, 1\}$
- **Model:** $P(x|\theta) = \theta^x(1 - \theta)^{1-x}$
 - $P(x|\theta) = \begin{cases} 1 - \theta. & \text{for } x = 0 \\ \theta. & \text{for } x = 1 \end{cases}$



MLE of Biased Coin

- Likelihood: $P(D|\theta) = \theta^{n_{head}}(1 - \theta)^{n_{tail}}$
 - n_{head} : number of heads, n_{tail} : number of tails

- Log-likelihood:

$$\begin{aligned} l(\theta; D) &= \log P(D|\theta) = \log \theta^{n_{head}}(1 - \theta)^{n_{tail}} \\ &= n_{head} \log \theta + (n - n_{head}) \log(1 - \theta) \end{aligned}$$

- Maximize $l(\theta; D)$ w.r.t. θ . We can set the derivative w.r.t. θ to be 0.

$$\frac{\partial l}{\partial \theta} = \frac{n_{head}}{\theta} - \frac{(n - n_{head})}{1 - \theta} = 0$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{n_{head}}{n} \text{ or } \hat{\theta}_{MLE} = \frac{1}{n} \sum_{i=1}^n x^i$$

Example: Gaussian Distribution

- Univariate Gaussian distribution in $\mathbb{R}$

$$P(x|\mu, \sigma) = \frac{1}{(2\pi)^{\frac{1}{2}}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x - \mu)^2\right)$$

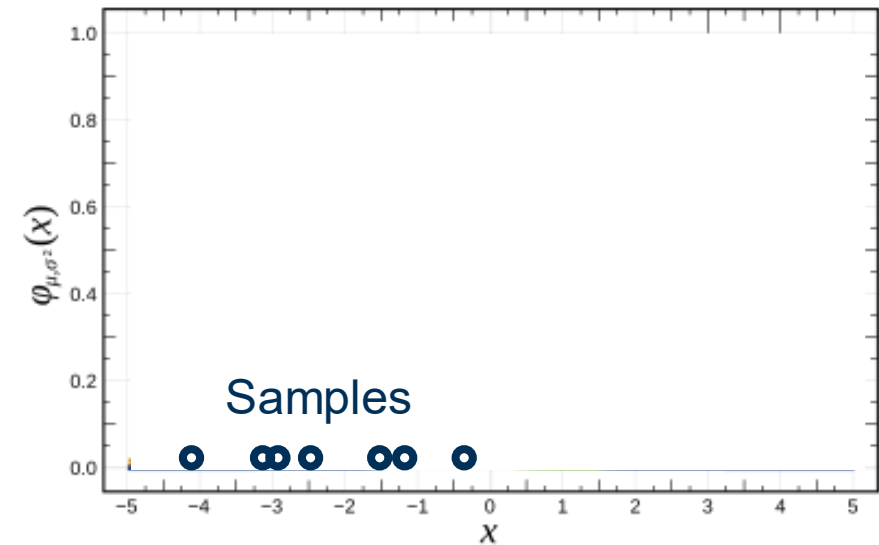
- Need to estimate two sets of parameters μ, σ

- Given n i.i.d. samples

$$D = \{x^1, x^2, \dots, x^n\}, \quad x^i \in \mathbb{R}$$

- Likelihood of one data point:

$$P(x|\mu, \sigma) \propto \exp\left(-\frac{1}{2\sigma^2}(x^i - \mu)^2\right)$$



MLE for Gaussian Distribution

- Objective function, log likelihood

$$\begin{aligned} l(\mu, \sigma; D) &= \log \prod_{i=1}^n \frac{1}{(2\pi)^{\frac{1}{2}} \sigma} \exp \left(-\frac{1}{2\sigma^2} (x^i - \mu)^2 \right) \\ &= -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 - \sum_{i=1}^n \frac{(x^i - \mu)^2}{2\sigma^2} \end{aligned}$$

- Maximize $l(\mu, \sigma; D)$ with respect to μ, σ . Set derivatives w.r.t. μ, σ^2 to be 0

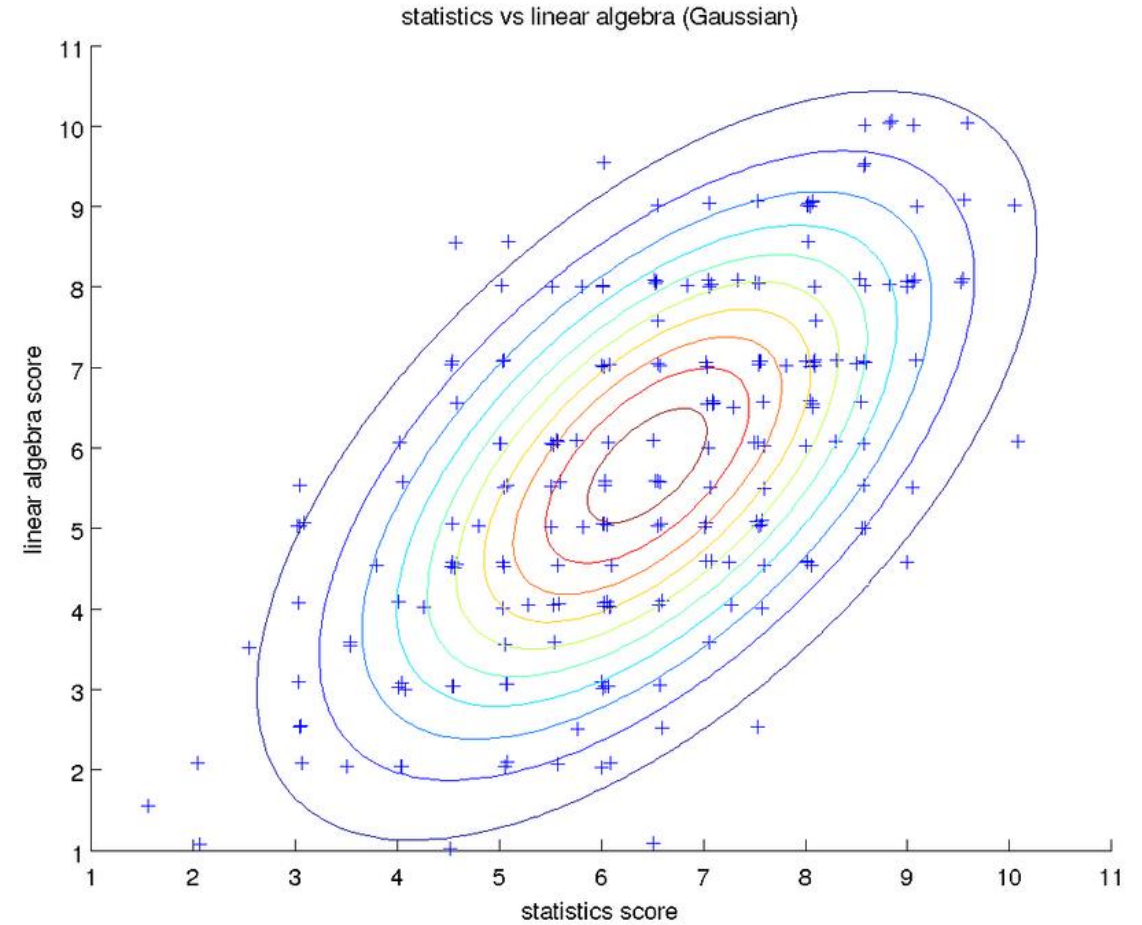
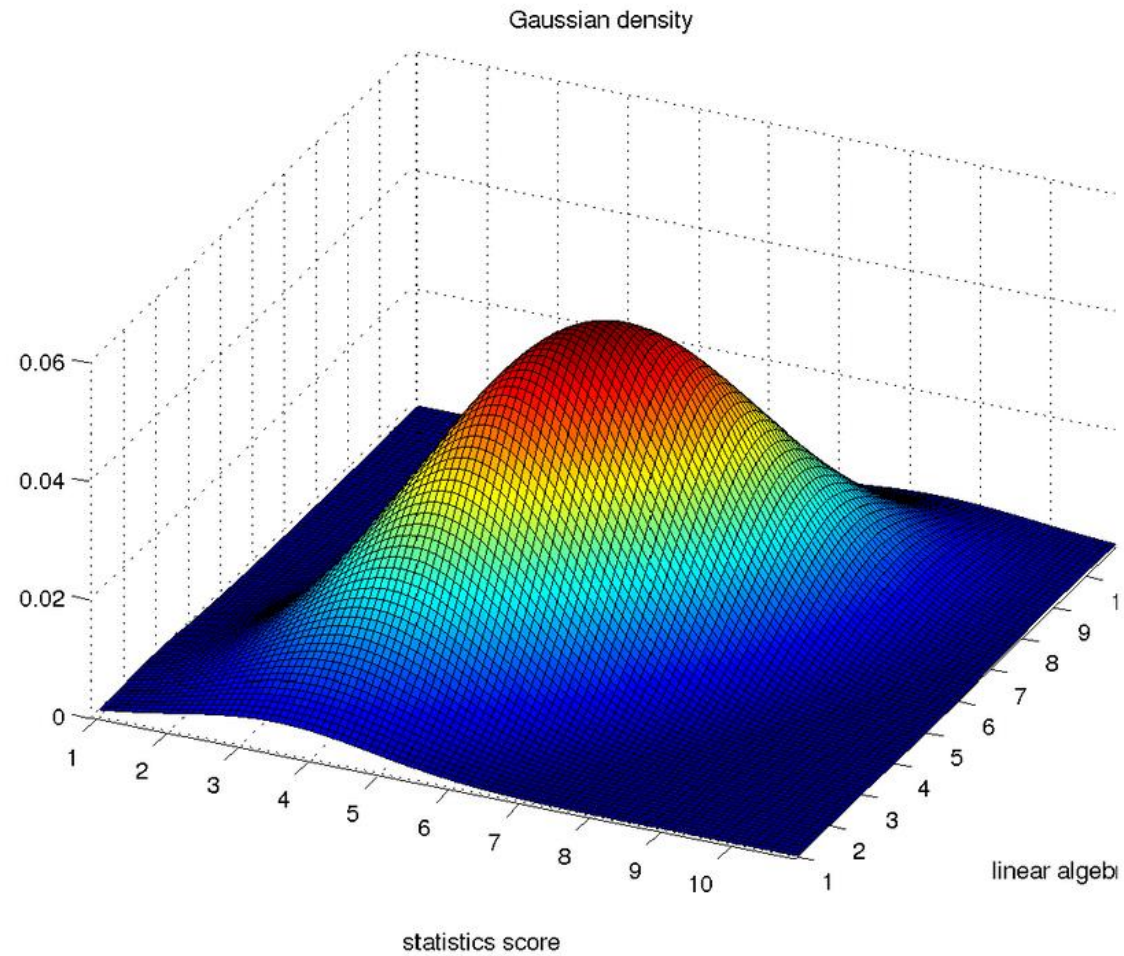
$$\frac{\partial l}{\partial \mu} = 0, \quad \frac{\partial l}{\partial \sigma^2} = 0$$

MLE for Gaussian Distribution

$$\frac{\partial l}{\partial \mu} = \sum_{i=1}^n \frac{x^i - \mu}{\sigma^2} = 0$$
$$\Rightarrow \sum_{i=1}^n x^i = n\mu \quad \Rightarrow \mu = \frac{1}{n} \sum_{i=1}^n x^i$$

$$\frac{\partial l}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x^i - \mu)^2 = 0$$
$$\Rightarrow \sum_{i=1}^n (x^i - \mu)^2 = n\sigma^2. \quad \Rightarrow \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x^i - \mu)^2$$

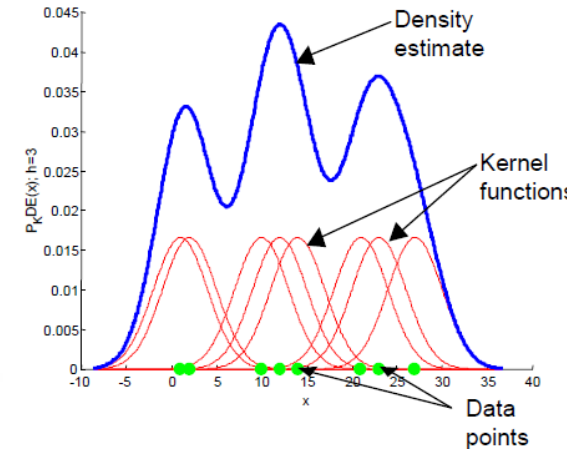
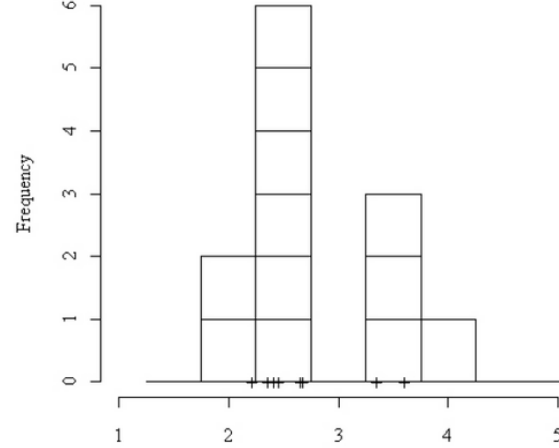
Density Example



Nonparametric Models

Nonparametric Models

- E.g., histogram
- E.g., kernel density estimator
- What are nonparametric models?
 - “Nonparametric” does **NOT** mean there are no parameters
 - Nonparametric models can **NOT** be described by **a fixed number of parameters**
 - One can think of there are many many (infinite) parameters



1-D Histogram

- One of the simplest nonparametric density estimator

- Given n i.i.d. samples $D = \{x^1, x^2, \dots, x^n\}, x^i \in [0,1)$

- Split $[0,1)$ into m bins

$$B_1 = \left[0, \frac{1}{m}\right), B_2 = \left[\frac{1}{m}, \frac{2}{m}\right), \dots, B_m = \left[\frac{m-1}{m}, 1\right)$$

- Count the number of points: c_1 points in B_1 , c_2 points in $B_2, \dots$

- For a new test point x

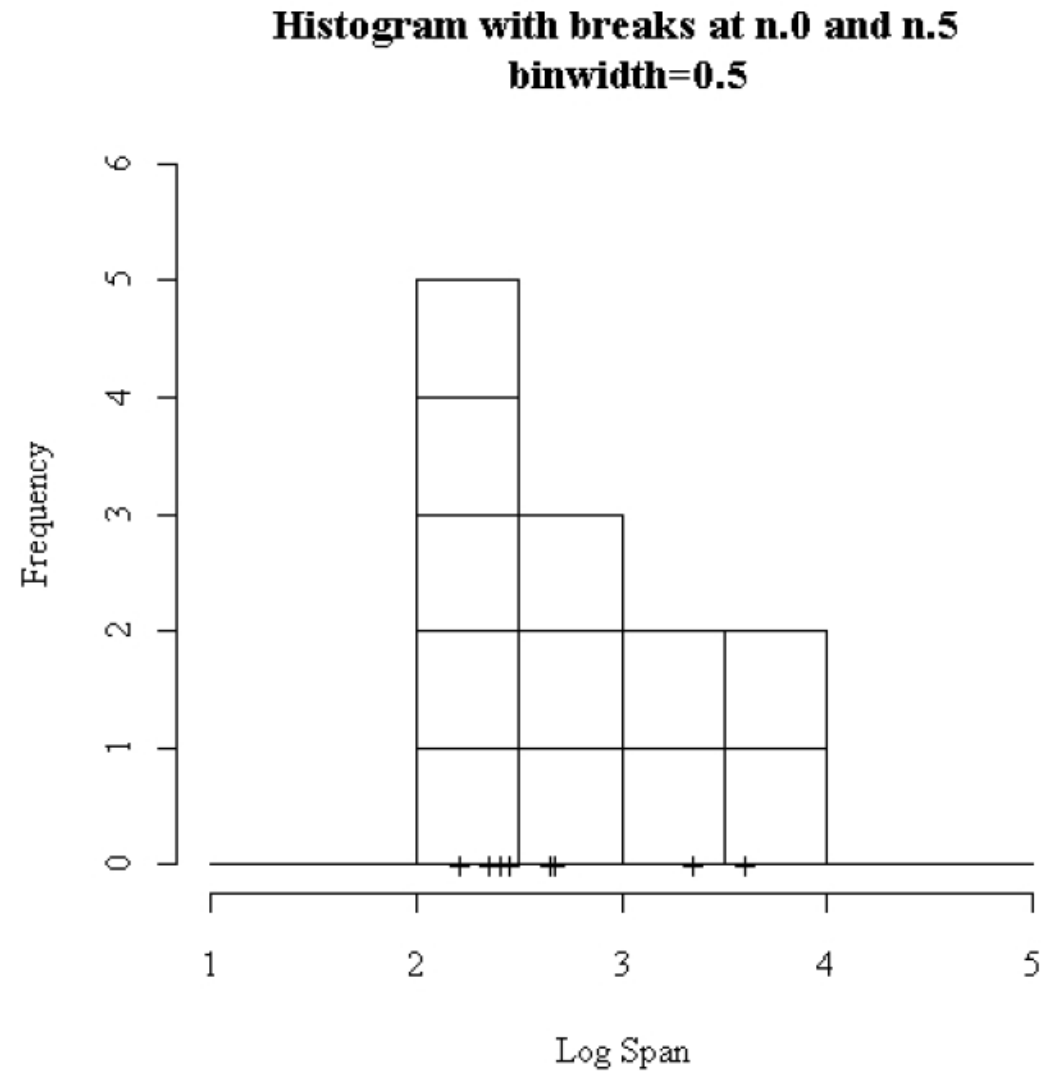
$$p(x) = \sum_{j=1}^m \frac{mc_j}{n} I(x \in B_j) \quad (\text{probability density function})$$

Why is Histogram Valid?

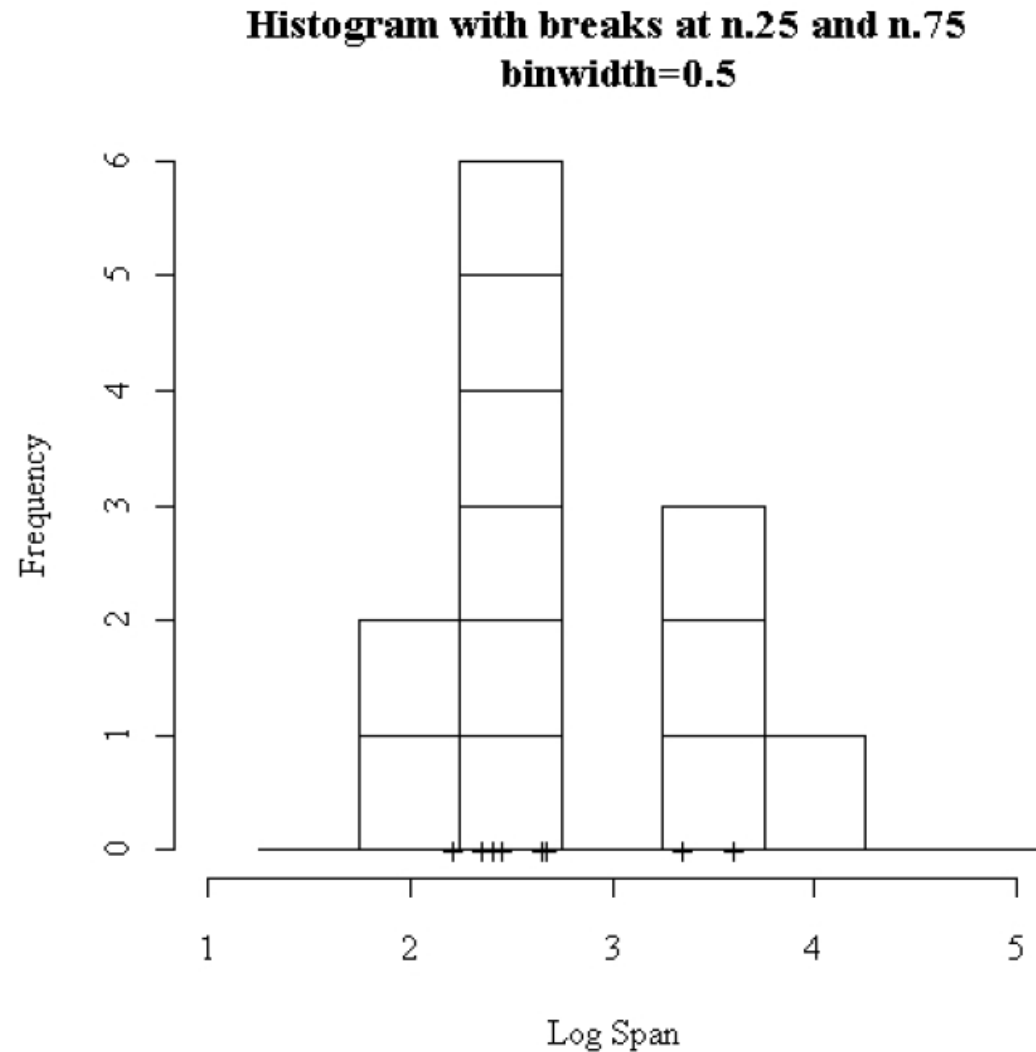
- Requirement for density $p(x)$
 - $p(x) \geq 0, \quad \int_{\Omega} p(x)dx = 1$
- For histogram,

$$\begin{aligned}\int_{\Omega} p(x)dx &= \int_{[0,1)} \sum_{j=1}^m \frac{mc_j}{n} I(x \in B_j) dx \\ &= \sum_{j=1}^m \int_{\left[\frac{j-1}{m}, \frac{j}{m}\right)} \frac{mc_j}{n} dx \\ &= \sum_{j=1}^m \frac{c_j}{n} = 1\end{aligned}$$

Output Depends on Where You Put the Bins



Output Depends on Where You Put the Bins



Higher Dimensional Histogram

- Given n i.i.d. samples $D = \{x^1, x^2, \dots, x^n\}$, $x^i \in [0,1)^d$

- Split $[0,1)^d$ evenly into m^d bins

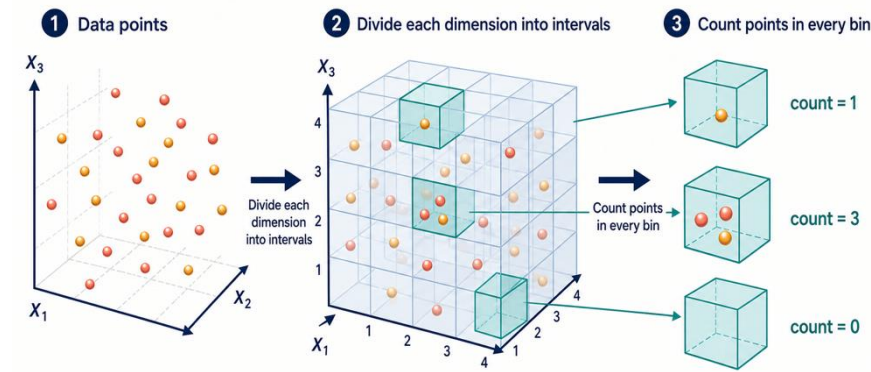
$$B_1 = \left[0, \frac{1}{m}\right) \times \left[0, \frac{1}{m}\right) \times \dots \times \left[0, \frac{1}{m}\right),$$

$$B_2 = \left[\frac{1}{m}, \frac{2}{m}\right) \times \left[0, \frac{1}{m}\right) \times \dots \times \left[0, \frac{1}{m}\right),$$

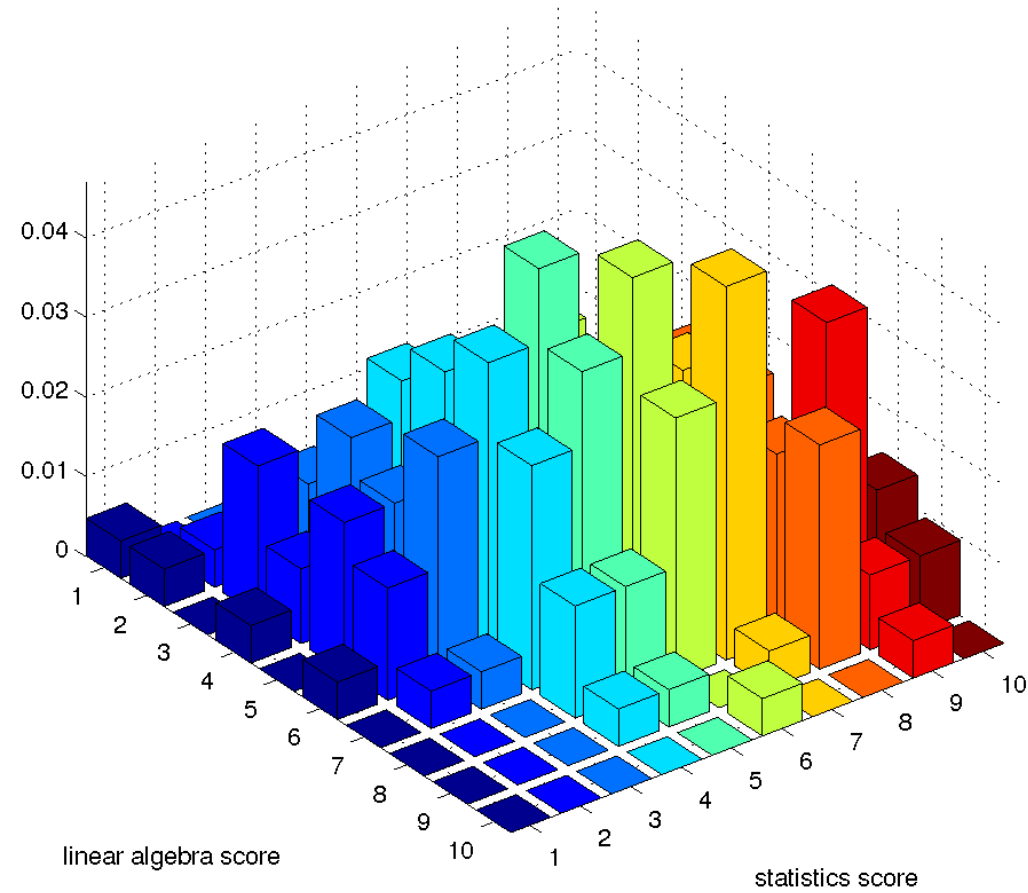
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$$B_{m^d} = \left[\frac{m-1}{m}, 1\right) \times \left[\frac{m-1}{m}, 1\right) \times \dots \times \left[\frac{m-1}{m}, 1\right)$$

- Bin size is $h = \frac{1}{m}$

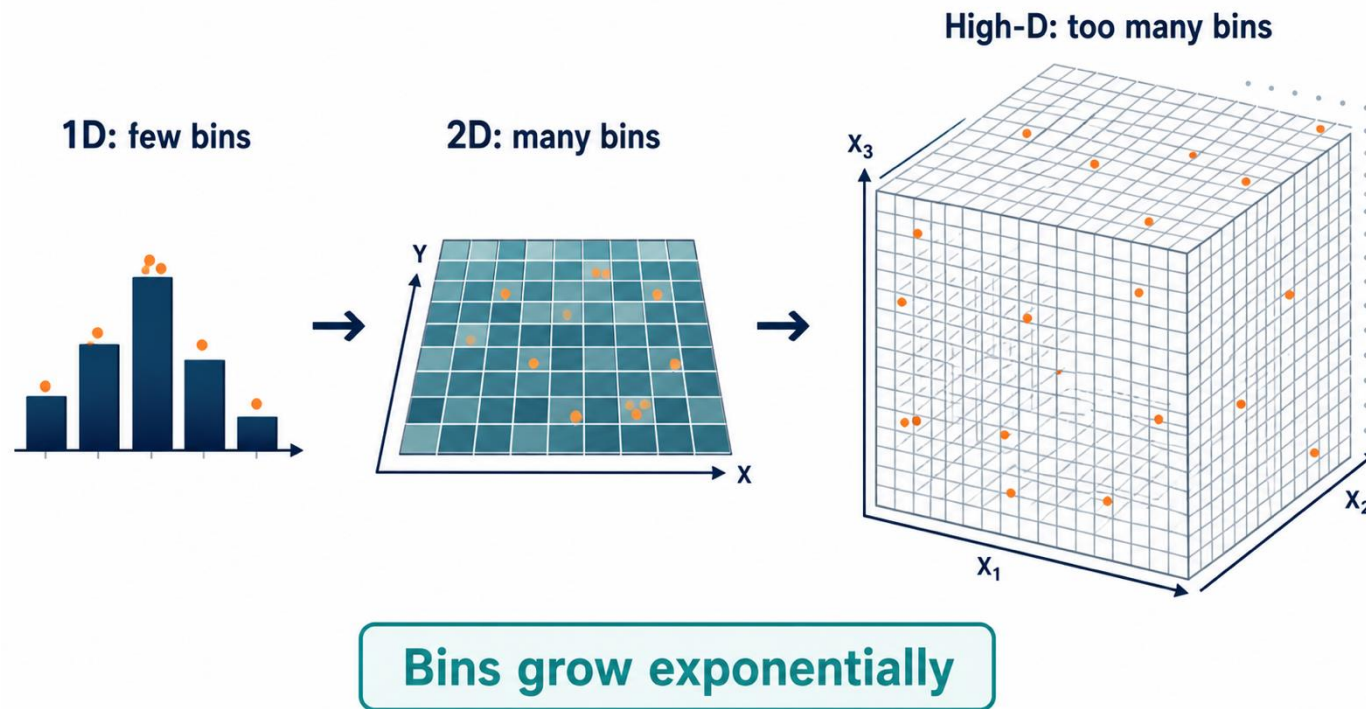


Class Scores



Computation Consideration

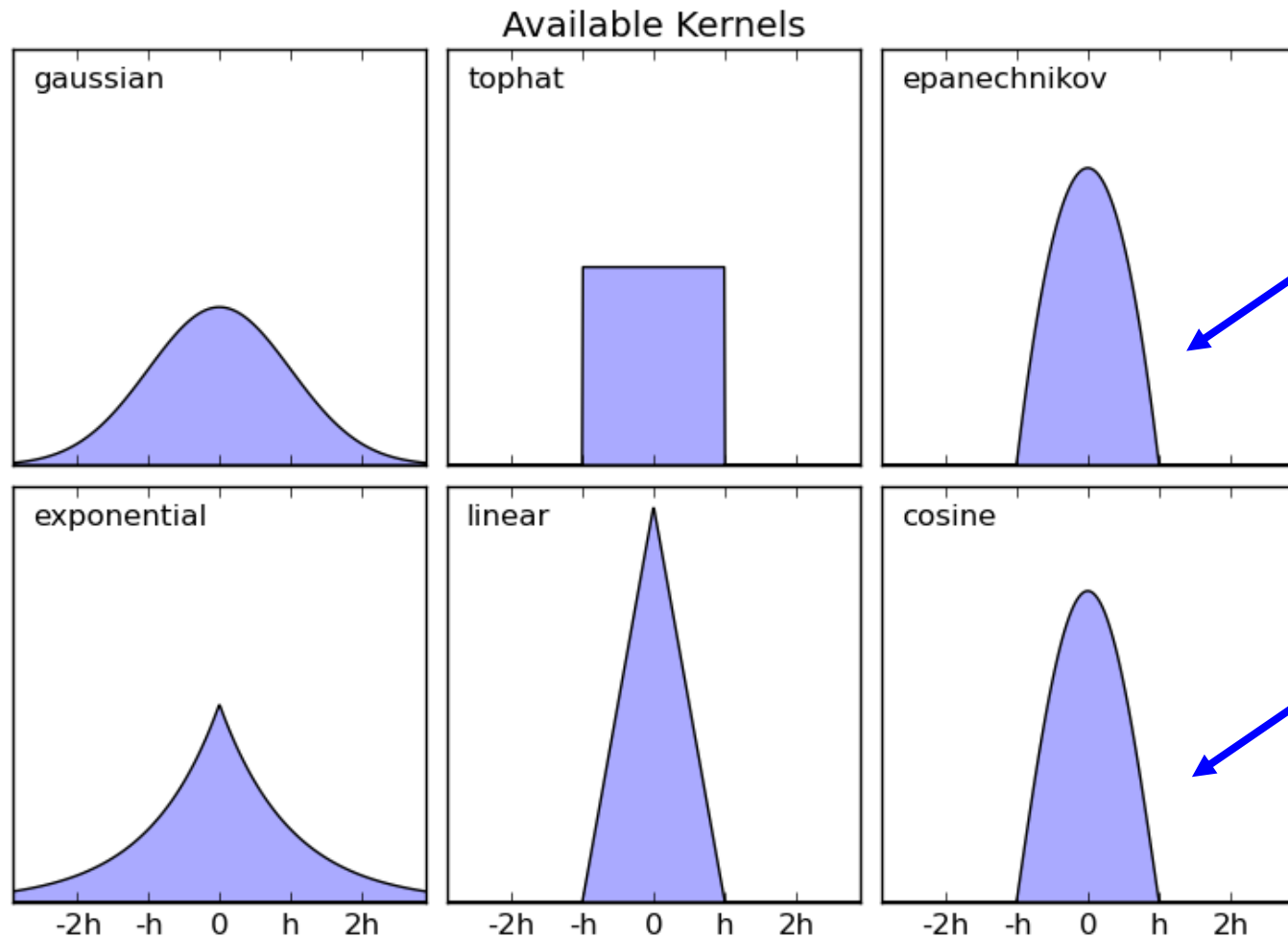
- **Problem:** too many bins! Not good for high dimensional data
 - If m^d is larger than the number of samples n , most bins are empty



Kernel Density Estimation

Kernel Functions

- Have you ever wondered why is histogram **rectangle**?



$$K(u) = \frac{3}{4}(1 - u^2)I(|u| \leq 1)$$

$$K(u) = \frac{\pi}{4} \cos\left(\frac{\pi}{2}u\right) I(|u| \leq 1)$$

Kernel Density Estimation

- Kernel density estimator

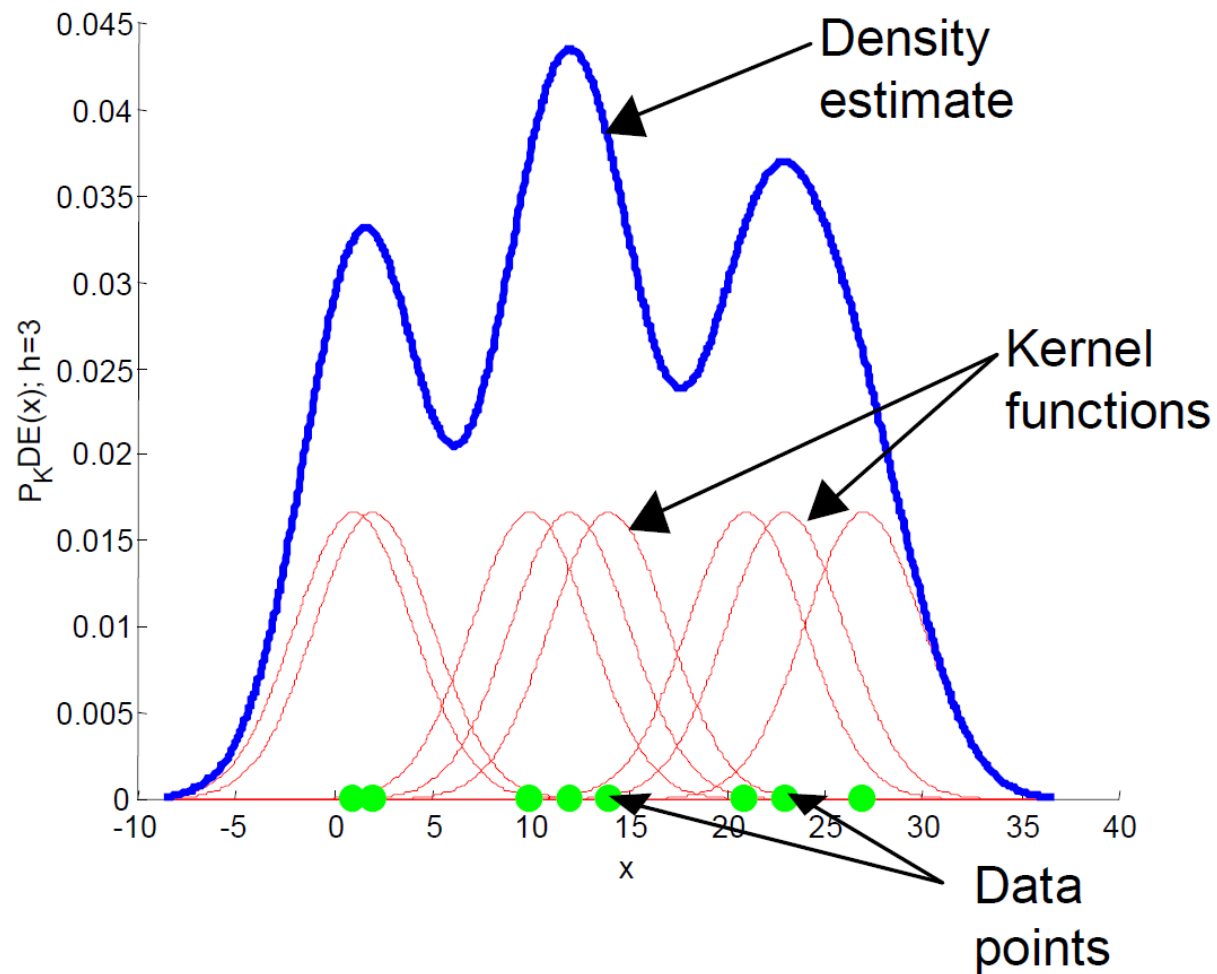
$$p(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$

- Smoothing kernel function

- $K(u) \geq 0$
- $\int K(u)du = 1$
- $\int uK(u)du = 0$
- $\int u^2K(u)du < \infty$

- An example: Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$

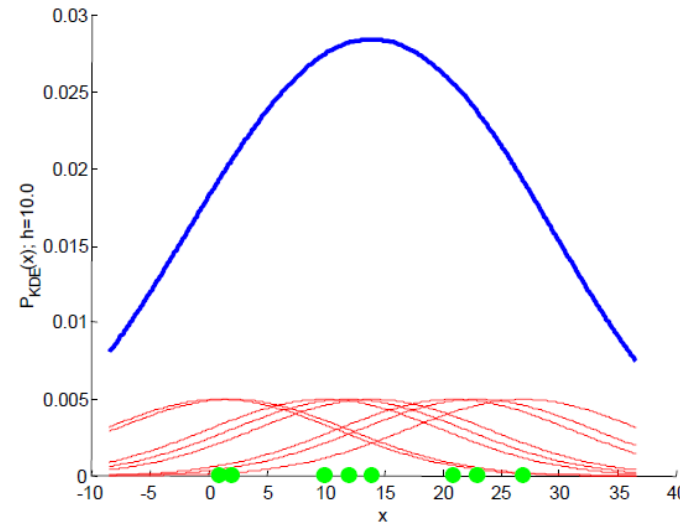
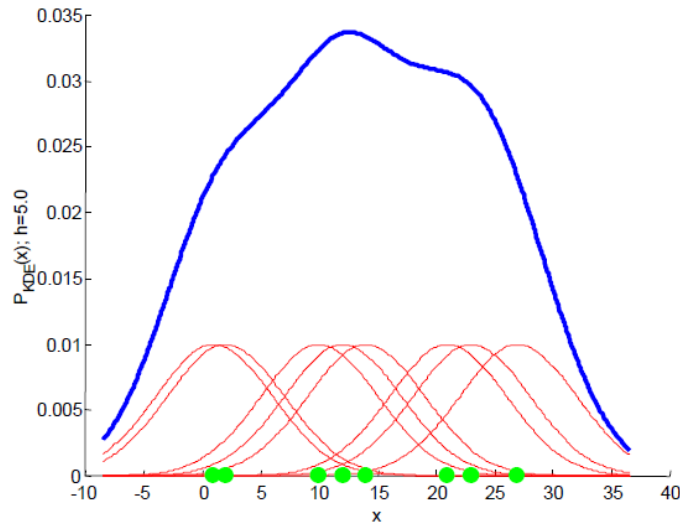
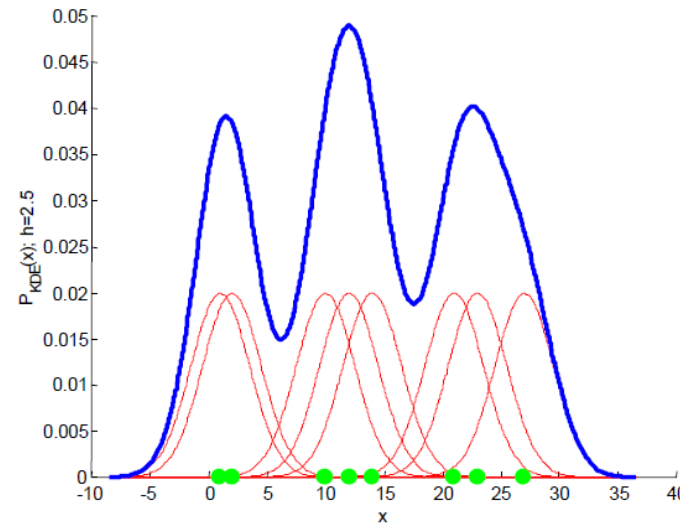
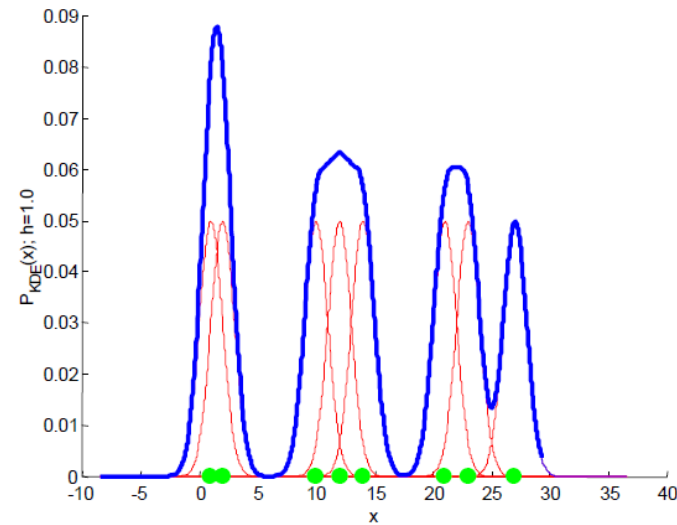
Example



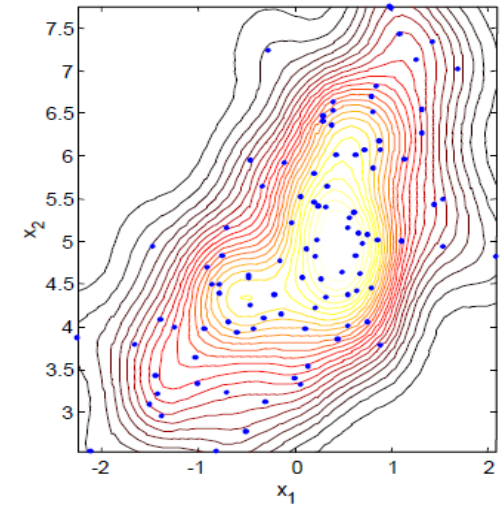
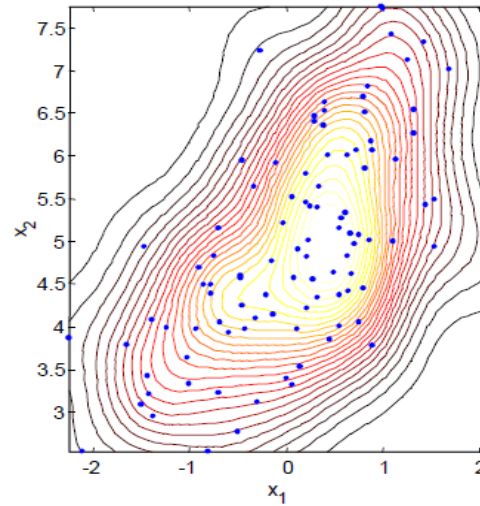
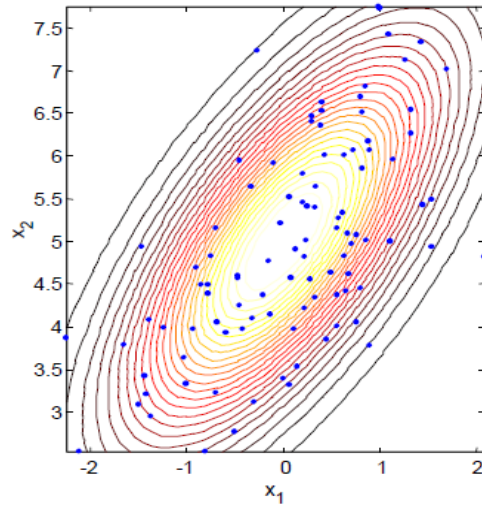
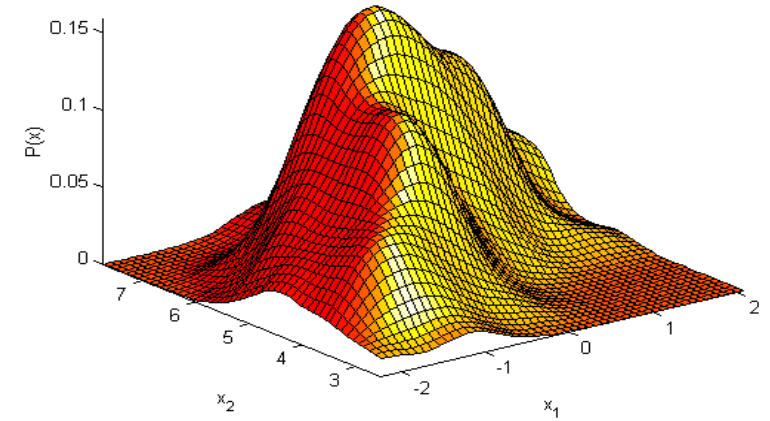
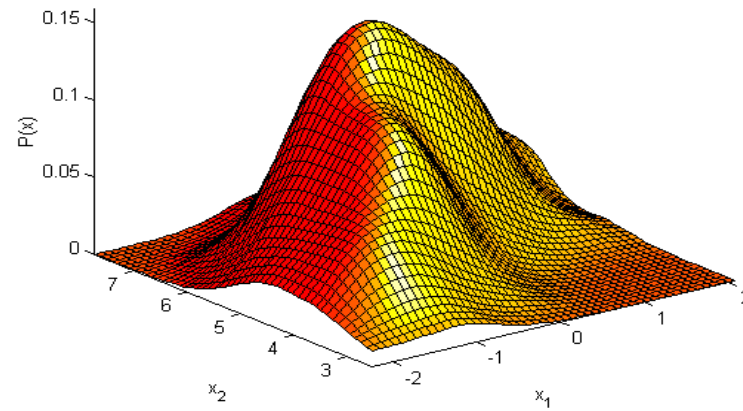
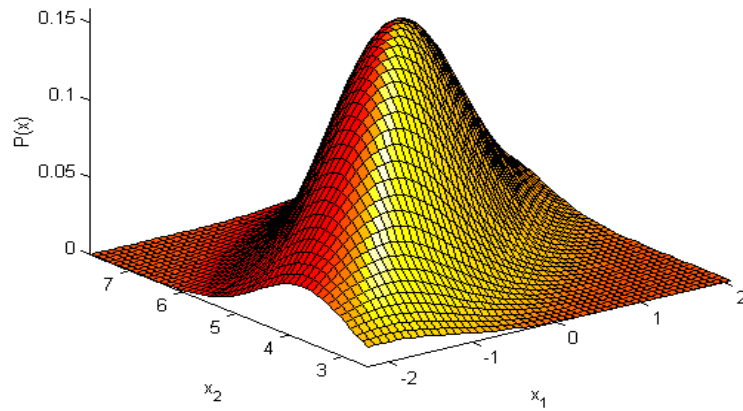
$$p(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$

Effect of the Kernel Bandwidth h

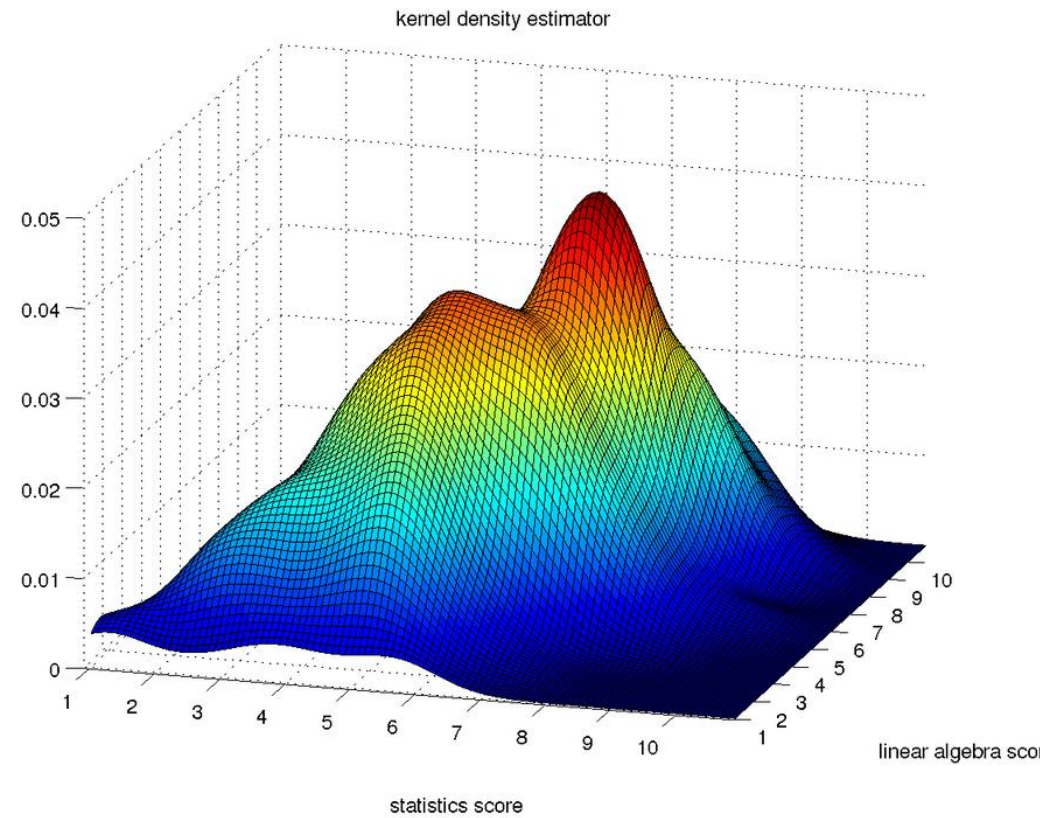
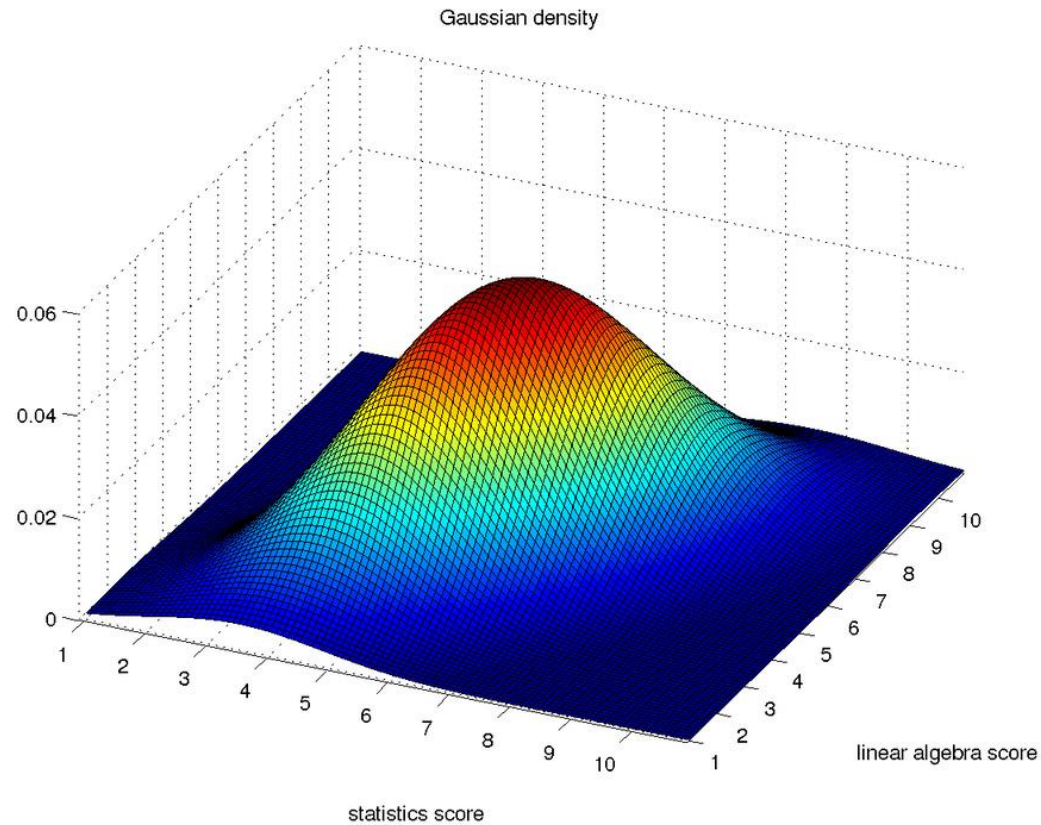
$$p(x) = \frac{1}{n} \sum_{i=1}^n \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$



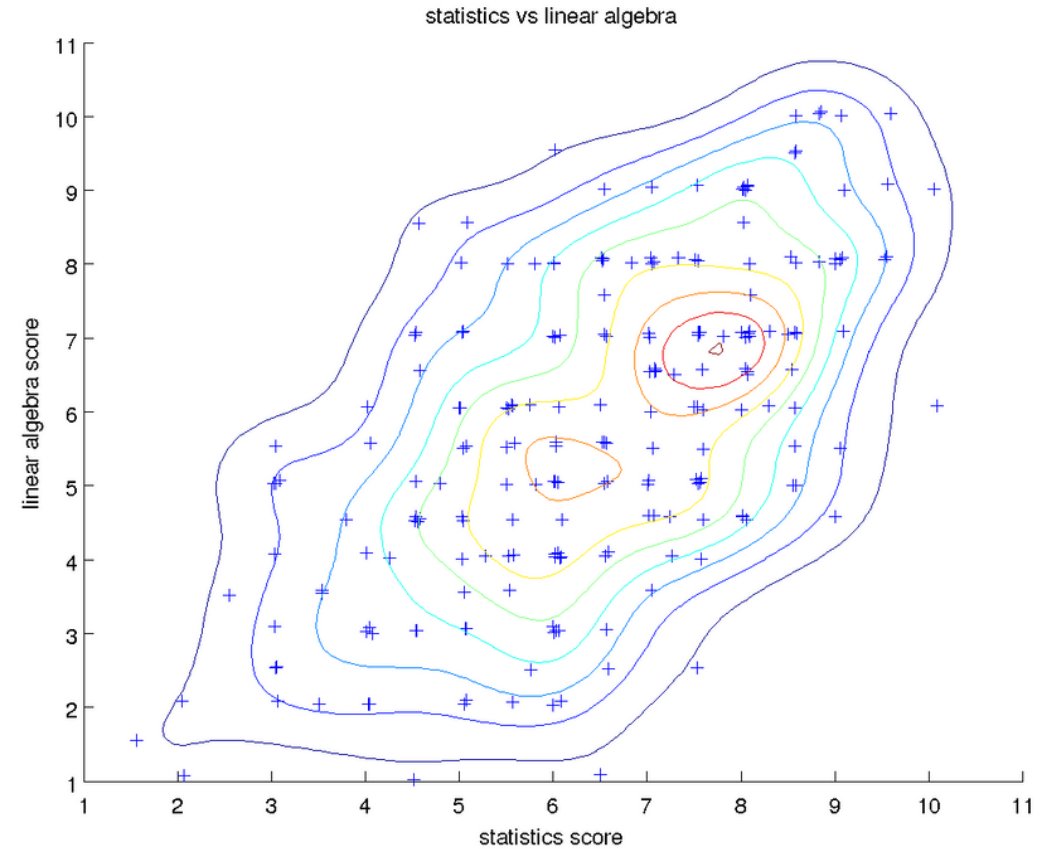
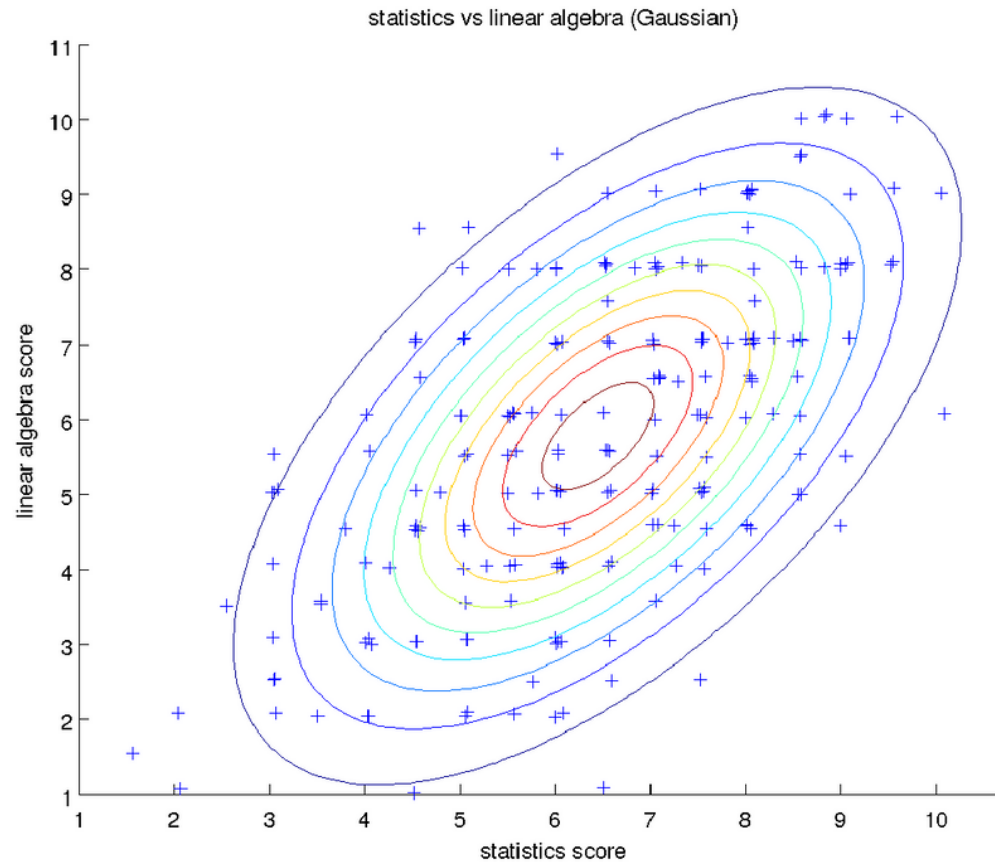
Two-dimensional Example



Parametric v.s. Nonparametric



Parametric v.s. Nonparametric



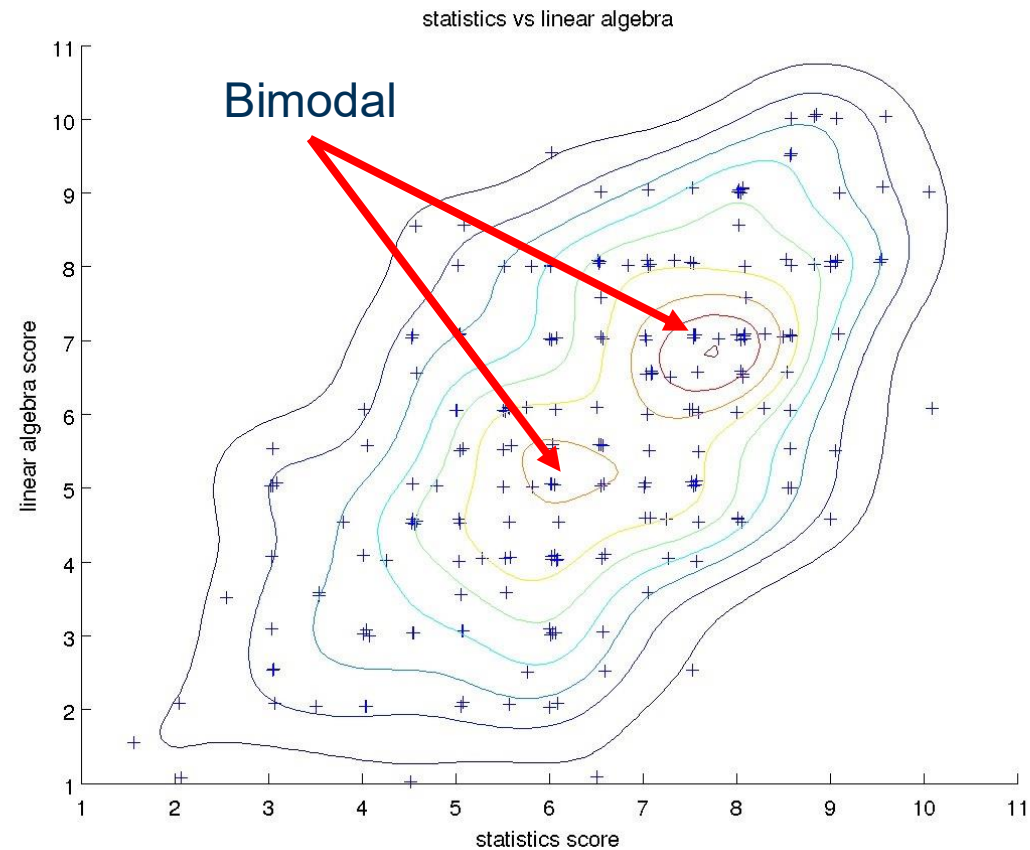
Parametric v.s. Nonparametric

- **Parametric** models rely on strong (simplistic) distributional assumptions
 - **Nonparametric** models place mild assumptions on the data distribution and provide good models for complex data
-
- **Parametric** models (once fitted) are much more efficient in terms of storage and computation.
 - **Nonparametric** models requires storing and computing with the entire data set.

Gaussian Mixture Models

Density Estimation

- Can we have a more flexible model than Gaussian, but less parameter than KDE?



Gaussian Mixture Model

- A density model $p(X)$ may be multi-modal
 - Model it as a mixture of uni-modal distributions (e.g., Gaussians)

$$\mathcal{N}(X|\mu_k, \Sigma_k) := \frac{1}{|\Sigma_k|^{\frac{1}{2}}(2\pi)^{\frac{d}{2}}} \exp\left(-\frac{1}{2}(X - \mu_k)^\top \Sigma_k^{-1}(X - \mu_k)\right)$$

- Consider a mixture of K Gaussians

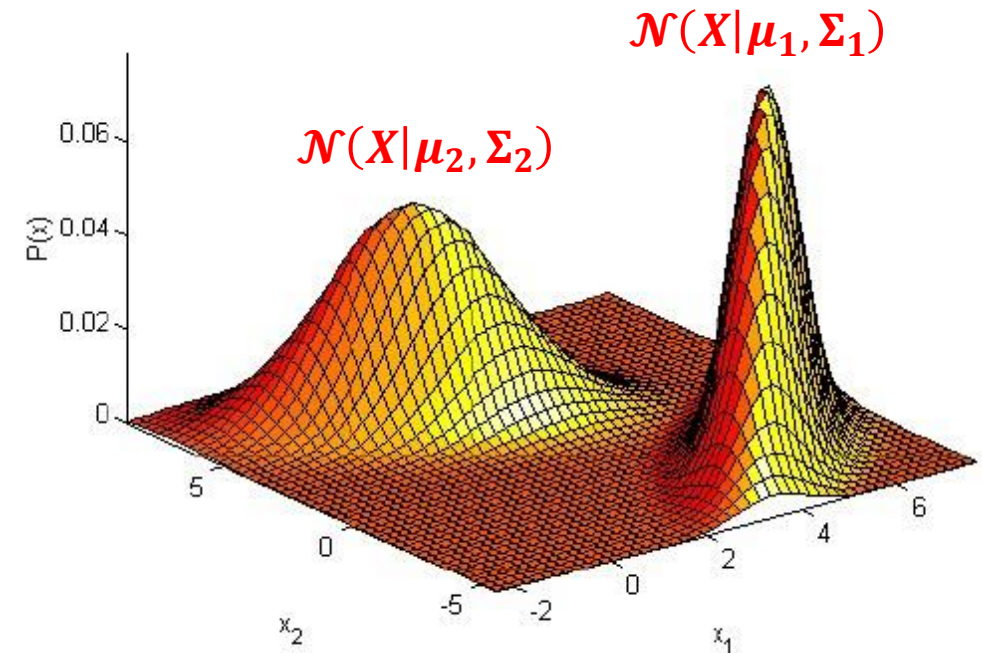
- $p(X) := \sum_{k=1}^K \pi_k \mathcal{N}(X|\mu_k, \Sigma_k)$

Mixing proportion

Mixing components

- **Question:** parametric or nonparametric?

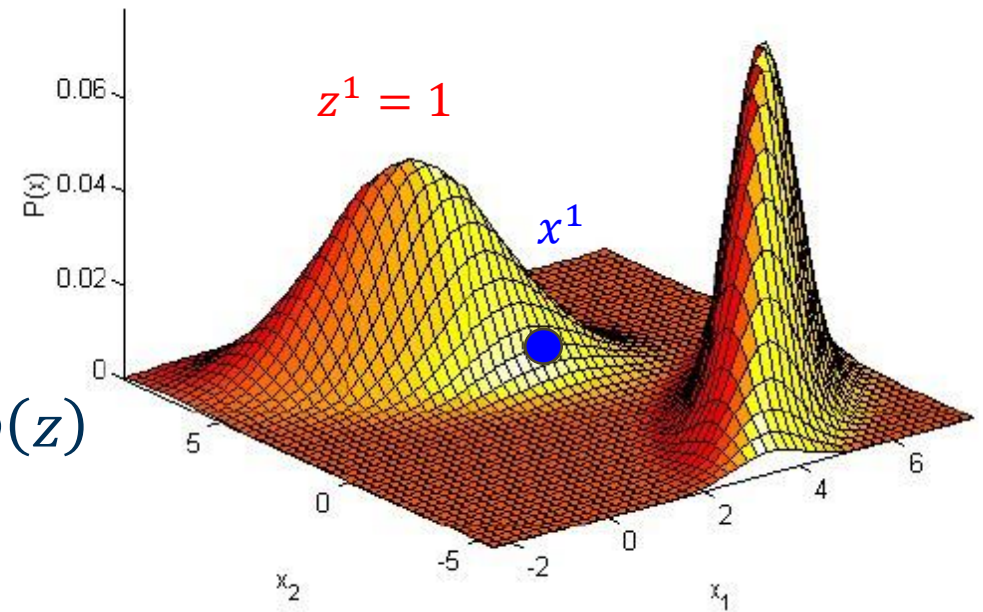
- Learn $\pi_k \in (0,1), \mu_k, \Sigma_k$



A Generative Process for Data Points

- Given $p(X) := \sum_{k=1}^K \pi_k \mathcal{N}(X|\mu_k, \Sigma_k)$, how to generate a data point $x^i \sim p$?
 - Randomly choose a mixture component, $z^i \in \{1, 2, \dots, K\}$, with probability π_{z^i}
 - Sample x^i from the corresponding Gaussian distribution $\mathcal{N}(X|\mu_{z^i}, \Sigma_{z^i})$
- Joint distribution $p(x, z)$ over x and z :
$$p(x, z) = \pi_z \mathcal{N}(X|\mu_z, \Sigma_z)$$
- Marginal distribution $p(x)$

$$p(x) = \sum_{z=1}^K p(x, z) = \sum_{z=1}^K p(x|z)p(z)$$



Learning the Parameters

- We know how to sample. But how to learn the parameters?
- **Maximum likelihood estimation (MLE)** by letting $\theta = (\pi_k, \mu_k, \Sigma_k)_{k=1,2,\dots,K}$:

$$\theta^* = \operatorname{argmax}_{\theta} l(\theta; D) := \log P(D|\theta)$$

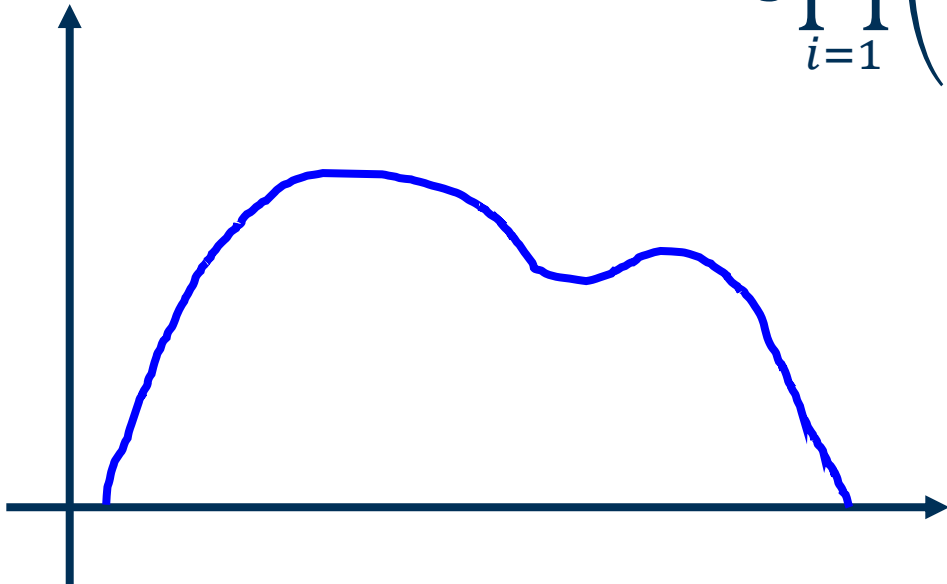
- Use our generative process

$$\begin{aligned} l(\theta; D) &= \log \prod_{i=1}^n p(x^i | \theta) \\ &= \log \prod_{i=1}^n \left(\sum_{k=1}^K p(x^i, z^i = k | \theta) \right) \\ &= \log \prod_{i=1}^n \left(\sum_{k=1}^K p(x | \mu_k, \Sigma_k) p(z^i = k | \pi) \right) \end{aligned}$$

Why is Learning Hard?

- The likelihood of the data becomes:

$$l(\theta; D) = \log \prod_{i=1}^n \left(\sum_{k=1}^K p(x|\mu_k, \Sigma_k) p(z^i = k|\pi) \right)$$
$$= \log \prod_{i=1}^n \left(\sum_{k=1}^K \pi_k \mathcal{N}(x|\mu_k, \Sigma_k) \right)$$



**Nonconvex!
Difficult!**

Next Week: EM (Expectation-Maximization) Algorithm

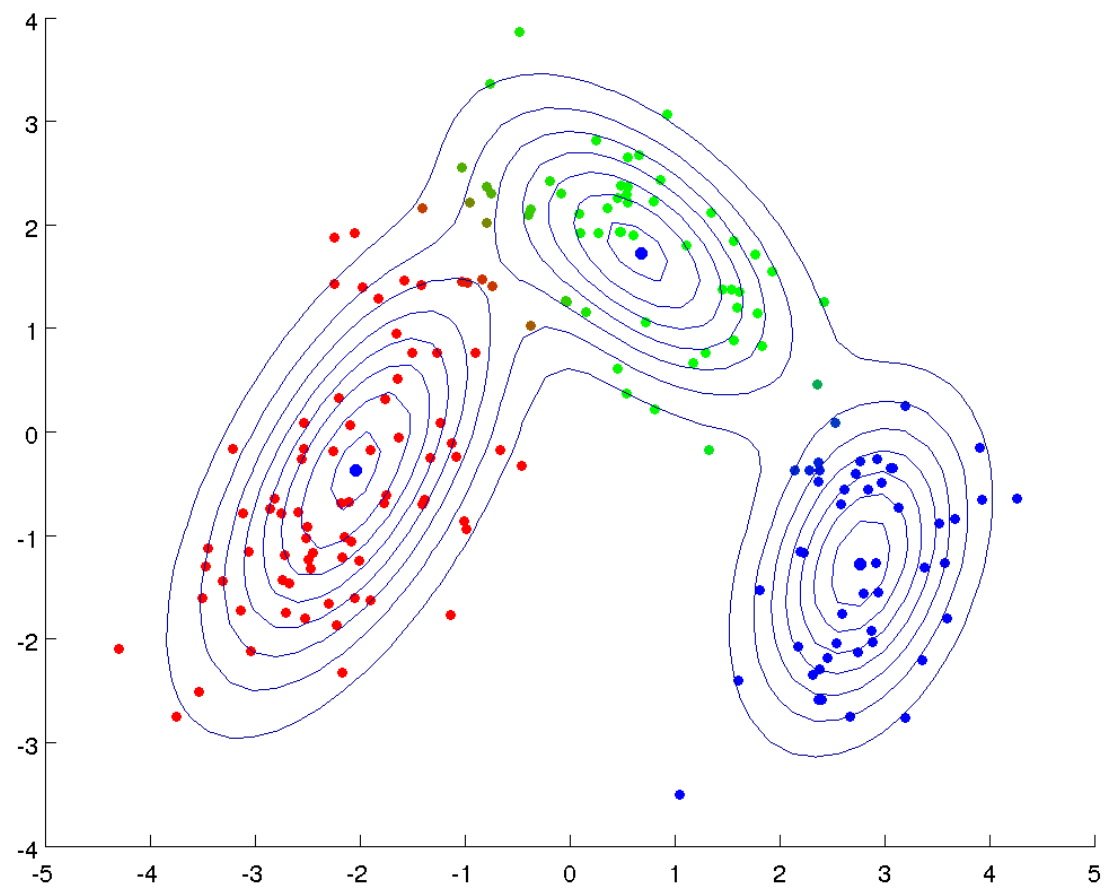
- Associate each data and each component with a τ_k^i
- Initialized $(\pi_k, \mu_k, \Sigma_k), k = 1, 2, \dots, K$
- Iterate the following two steps until convergence:
 - **Expectation step (E-step, Gaussian assignment)**: update τ_k^i given the current (π_k, μ_k, Σ_k)

$$\tau_k^i = p(z^i = k \mid D, \mu, \Sigma) = \frac{\pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(x \mid \mu_{k'}, \Sigma_{k'})}, \quad \forall k \in \{1, 2, \dots, K\}; i \in \{1, 2, \dots, n\}$$

- **Maximization step (M-step, Gaussian adjustment)**: update (π_k, μ_k, Σ_k) given τ_k^i

$$\begin{aligned} \pi_k &= \frac{\sum_i \tau_k^i}{n}, & \mu_k &= \frac{\sum_i \tau_k^i x^i}{\sum_i \tau_k^i} \\ \Sigma_k &= \frac{\sum_i \tau_k^i (x^i - \mu_k)(x^i - \mu_k)^\top}{\sum_i \tau_k^i}, & \forall k &\in \{1, 2, \dots, K\} \end{aligned}$$

Demo



Announcement

- No class on **Sep 7th**. Happy long weekend!
- Hw1 will be released on **Sep 6th (Sunday)** and due on **Sep 20th (Sunday)**. Please start working on it earlier!